

1. **MS:** 8.1 (a-h)2. **Additional Questions:**

- (a) A firm has two different plants. Let the cost at plant 1 be given by $c(y_1) = 5y_1^2 + 100y_1$, and the cost at plant 2 by $c(y_2) = 500y_2$. Suppose the firm wants to produce $\bar{Q}$ units of total output.
- (a) Write out the total cost for the firm.
 - (b) Find the marginal cost of production at each plant.
 - (c) What is the total cost minimizing level of output at each plant (as a function of the total output)?
 - (d) Suppose someone incorrectly suggests that the firm produce 21 units using the first plant. How should the firm adjust their production among the plants to correct this error.
- (b) *Silky Smoothies* production process can be described in the following way $F(L, K) = L^{\frac{1}{4}}K^{\frac{1}{2}}$. Where L and K broadly define capital and labor inputs used in the production of smoothies. Suppose K is fixed at 4 units. Graph *Silky Smoothies* total product of labor curve. For what values of L does this firm exhibit diminishing returns to labor? In practice, why would a firm exhibit diminishing returns to labor?
- (c) A firm faces the following demand curve: $P = 120 - 0.02Q$; where Q is weekly level of production and P is price, measured in cents per unit. The firm's cost function is given by $C = Q + 25,000$. Assume that the firm maximizes its profits. What is the level of production, price, and total profit per week?